DHRUV LAD

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Education

Veermata Jijabai Technological Institute (VJTI), Matunga — **Aug. 2024 – Present**
*B.Tech in Computer Engineering (Third Year) with a minor in Cybersecurity; CGPA: **7.89*** — *Mumbai, India*

Shri T.P. Bhatia College of Science, Kandivali — **Aug. 2022 – May 2024**
*MHT-CET: **99.97%ile** (Rank 90), Math: **100%ile** — HSC: **88.67%** — JEE Main: **97.34%ile*** — *Mumbai, India*

Relevant Coursework

- Object Oriented Programming
- Discrete Mathematics
- Operating Systems
- DBMS
- Data Structures and Algorithms
- Cybersecurity

Projects

FinSight | *Python, Sentence Transformers, FAISS, Streamlit, pandas, NumPy* — **Aug. 2026**

- Developed **temporal RAG** over **20 earnings-call transcripts** (**1,481 rows** → **1,544 chunks**; 700/120-word overlap) with **384-dim all-MiniLM-L6-v2** embeddings and **FAISS IndexFlatIP** ticker/quarter filtering.
- Integrated **gpt-oss-120b** via Groq (temperature=0) with context-constrained prompting for source-cited answers; achieved **100% Recall@3** and **0.90 MRR** on a 5-query evaluation set.

GestureFlow | *Python, PySide6, OpenCV, MediaPipe, PyAutoGUI, pytest, PyInstaller* — **May 2026**

- Built a real-time touchless **HCI desktop app** for Windows cursor control using **MediaPipe Hand Landmarker** (**21 landmarks/hand**) with deterministic landmark-based gesture recognition for click, drag, and scroll.
- Designed a **QThread** + **Qt Signals/Slots** pipeline decoupling capture, inference, and UI; stabilized input with **exponential smoothing**, hysteresis, and **3-frame** click debouncing; tested with **6 pytest modules**.

Voyexa | *React, Tailwind CSS, Spring Boot, PostgreSQL, Docker* — **Mar. 2026**

- Engineered a full-stack **AI travel planner** with a **2-stage Gemini** planner + validator prompt pipeline, **JSONB** persistence in PostgreSQL, **10-attempt** retries, API-key rotation, and exponential backoff.
- Implemented **JWT** auth (**BCrypt**, **24h / 7d** access/refresh tokens); integrated **Geoapify**, **Nominatim**, and **Pexels** APIs with **SSE** image streaming and **dnd-kit** itinerary editing.

Visual Semantic Encoder | *Python, PyTorch, TorchVision, NLTK, NumPy, Matplotlib, CUDA* — **Jan. 2026**

- Built an end-to-end **image captioning** system in PyTorch: custom **5-stage CNN encoder** (3→512 channels), **additive attention**, and **LSTM decoder** (256-dim embeddings, 512-dim hidden state).
- Trained on a **60K-image COCO subset** for **60 epochs** with **teacher forcing**, PAD-masked cross-entropy, and Adam on **CUDA**; evaluated with **BLEU-1/BLEU-4** and attention heatmaps.

Technical Skills

Core Concepts: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP)
Languages: C++, Python, Java, JavaScript, SQL, HTML/CSS
AI/ML & Data: PyTorch, TensorFlow, Scikit-learn, RAG, LLM Integration, Pandas, NumPy
Frameworks & Backend: React.js, Spring Boot, REST APIs, FastAPI
Developer Tools: Git/GitHub, Docker, VS Code, IntelliJ IDEA, Postman, Linux, Vercel, Render
Databases: MySQL, PostgreSQL, MongoDB, Supabase

Competitive Programming

- Solved **700+ problems** across LeetCode, CodeChef, and Codeforces in Data Structures and Algorithms.
- LeetCode: **Knight**, max **1981** | CodeChef: **2-Star**, max **1569** | Codeforces: **Newbie**, max **1069**.

Experience

Community of Coders, VJTI | *Inheritance Program, Apprentice* — **Nov. 2025 – Jan. 2026**

- Developed project-building and collaborative development skills through hands-on learning and peer collaboration.

Hacktoberfest 2025 | *Open Source Contribution* — **Oct. 2025**

- Delivered **6+ merged PRs** to diverse open-source repositories and earned the **Supercontributor Badge**.

Technovanza, VJTI | *Department Coordinator* — **Oct. 2024 – Oct. 2025**

- Coordinated departmental activities and responsibilities to support the planning and execution of college events.